



Parktown Boys' High School

Geography Department

Grade 12 Control Test

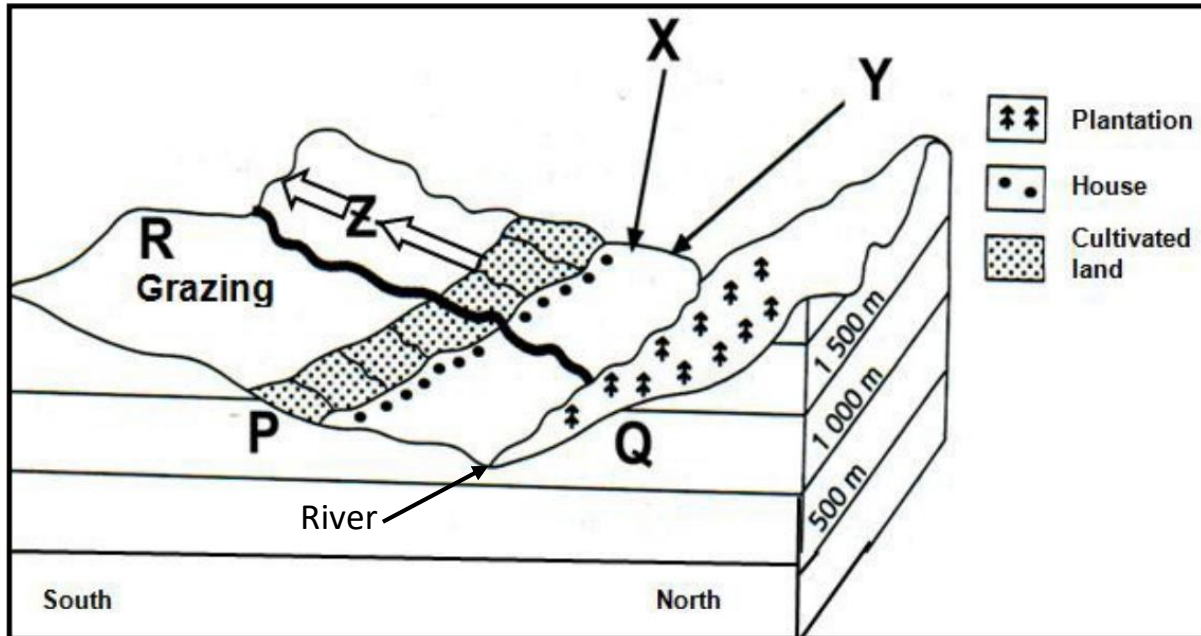
MEMO

INSTRUCTIONS AND INFORMATION

1. The question paper consists of FOUR questions.
2. ALL diagrams are included in the question paper.
3. Where possible, illustrate your answers with labelled diagrams.
4. Leave a line between subsections answered.
5. Number your answers correctly according to the numbering system used in this question paper.
6. Write neatly and legibly.

Question 1 – Short Questions [15]

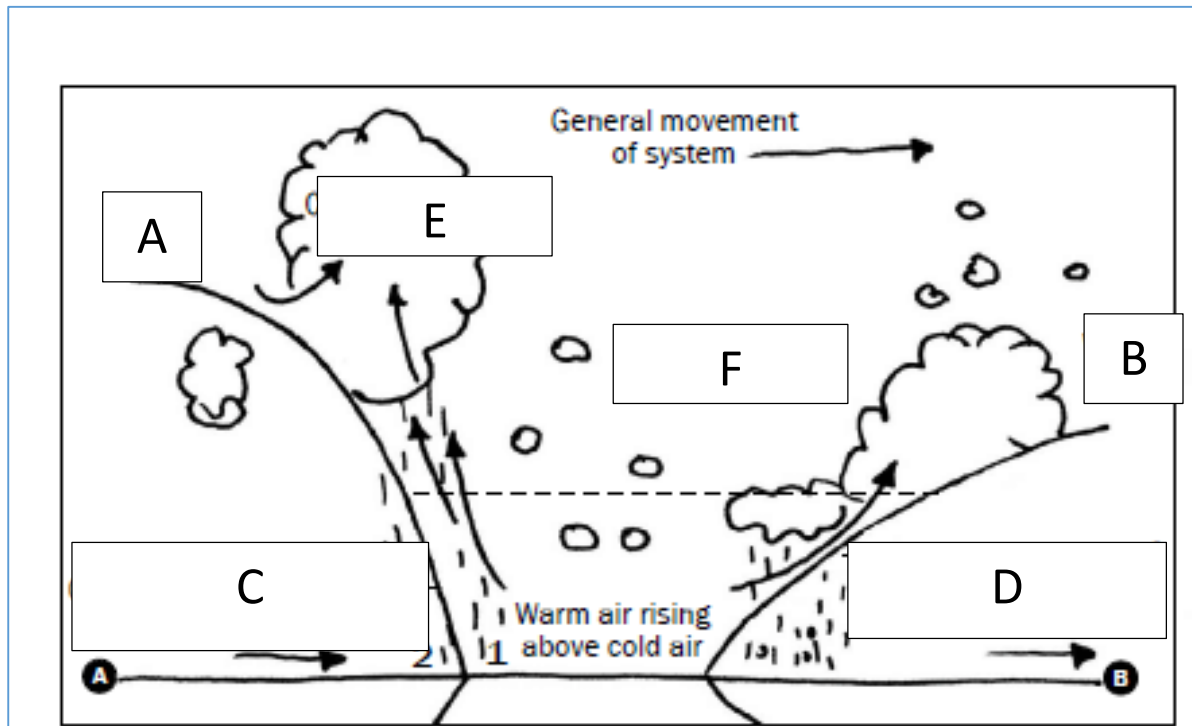
1.1. Refer to the diagram below showing a valley. Answer the questions that follow by choosing only the correct term in brackets. E.g. 1.9. Updog



- 1.1.1. Winds that travel along the valley floors and down the slopes during the night are called (anabatic/katabatic/upslope/downslope) winds.
- 1.1.2. Temperatures at (X/Y) will be warmer during the winter months.
- 1.1.3. (R/Q) would be north-facing slopes and therefore cooler in summer.
- 1.1.4. The plantation at Q would likely be growing (delicate, exotic fruit trees/robust, hardwood timber trees).
- 1.1.5. Houses at X benefit from (the inversion layer/cultivated land) during winter.
- 1.1.6. Inversion layers are created in winter when conditions are (clear and cool/windy/cloudy/warmer than normal).
- 1.1.7. The reason the area at Y has been left barren is due to the dangers of (frost pockets/pollution domes/urban heat island effect) in winter.
- 1.1.8. If this diagram is in the Southern Hemisphere, then the river is flowing in a (westerly/easterly) direction.

(8x1) (8)

1.2. Refer to the figure below and statements that follow. Select only the correct word in brackets in each statement to make the statement TRUE.



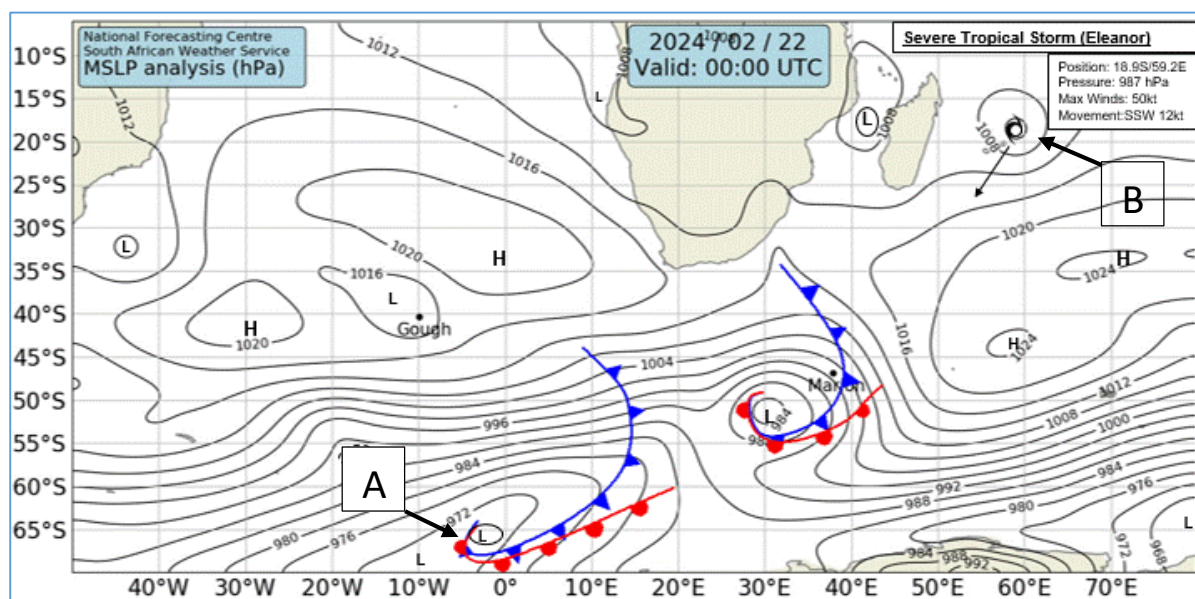
- 1.2.1. The front at A is the (warm / cold / occluded) front.
- 1.2.2. The temperatures of the air found at C, F and D are (cool, cold and warm / cold, warm and cool / warm, cold and cool) respectfully.
- 1.2.3. The symbol for the front found at B is (semi-circles / triangles / a combination of semi-circles and triangles).
- 1.2.4. The clouds that typically form at the front found at B are (cumulonimbus / stratus / cirrus) clouds.
- 1.2.5. The clouds that typically form at the front found at A are (cumulonimbus / stratus / cirrus) clouds.
- 1.2.6. The pressure found at the front labelled A is (low / high / no) pressure.
- 1.2.7. Once the front labelled A passes over a point, the pressure in that area will (increase / decrease / remain the same).

(7x1) (7)

1	Katabatic	1	Cold
2	X	2	Cold, warm and cool
3	R	3	Semi-circles
4	Hardwood	4	Stratus
5	Inversion	5	Cumulonimbus
6	Clear and Cool	6	Low
7	Frost pockets	7	Increase
8	Easterly		

Question 2 – Tropical Cyclones [15]

Refer to the synoptic weather map below and answer the questions that follow.



- 2.1. What season is being represented in the synoptic chart above? (1x1) (1)
Summer ✓
- 2.2. Provide TWO pieces of evidence to justify your answer to QUESTION 2.1. (2x1) (2)
[Any TWO]
Tropical Cyclone present, HP are far south, Date, Mid-lat cyclones are far south ✓✓
- 2.3. Name the TWO weather phenomena found at A and B respectively. (2x1) (2)

A – Mid-latitude cyclone ✓

B – Tropical cyclone ✓

- 2.4. Tropical Storm Eleanor is not yet a mature tropical cyclone. Fully describe TWO factors necessary for the formation of mature tropical cyclones. (2x2) (4)

[Any TWO – MUST be a discussion. Lists = ZERO]

Hot, moist air – Oceans hotter than 28 degrees ✓✓

Intense low pressure cell – Needs to be clear that this must be an intense low pressure cell. Disturbances in the Jet Stream lead to compounding effects of the LP ✓✓

Coriolis force – It is a spiralling storm, therefore needs to spiral. ✓✓

Latent heat – Release of latent heat keeps the energy in the cyclone. ✓✓

- 2.5. Although Tropical Storm Eleanor is not a fully-fledged cyclone, the people of Madagascar should still be on alert for the passing of this storm should its trajectory change. Describe ONE strategy that the government of Madagascar can implement to reduce the impact of this storm. (1x2) (2)

[Any ONE – Needs to be a description, i.e. WHY this will help people]

Prediction and Warning – Get people out ✓✓

Preparing the Community - Reduce financial and social impact ✓✓

Land Use Planning – Future protection ✓✓

Managing the Loss – Help recovery ✓✓

- 2.6. Tropical Storm Eleanor is predicted to miss Madagascar and head south where it will die out. Discuss why this storm will die out as it moves south. (2x2) (4)

[Accept reasonable answers]

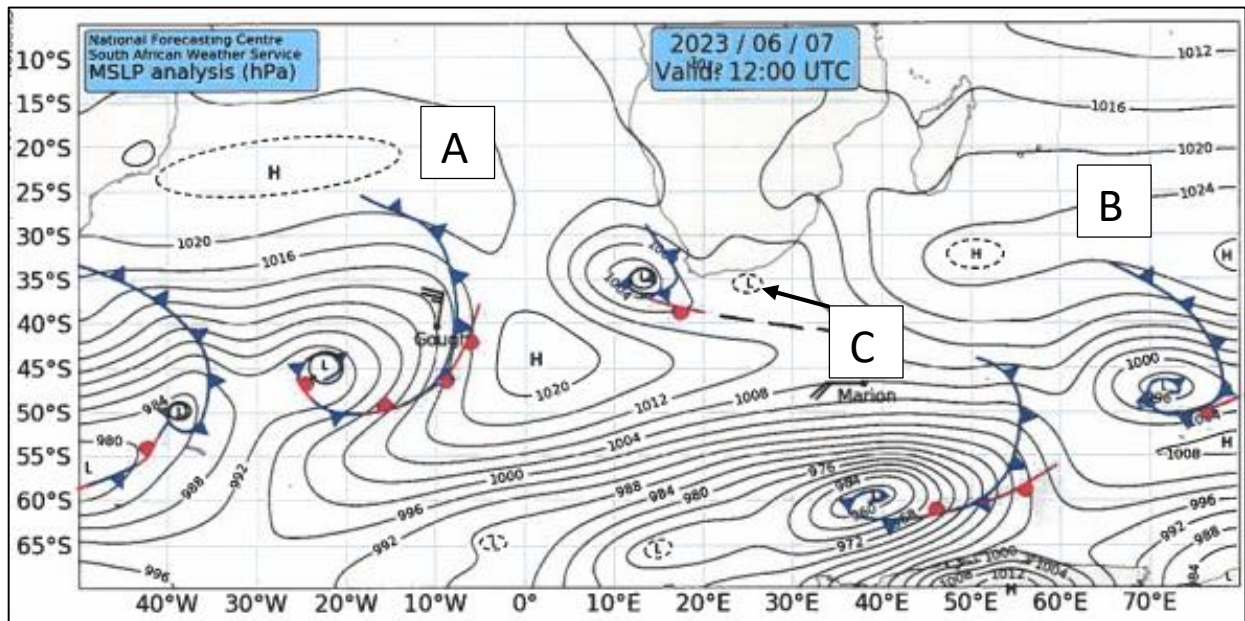
Enters cold water – pressure increases, no moisture ✓✓

Lack of evaporation leads to a lack of latent heat ✓✓

DO NOT ACCEPT FRICTION

Question 3 – Anti-cyclones [15]

Refer to the synoptic chart below and answer the questions that follow.



- 3.1. What season is being experienced in the above synoptic weather chart? (1x1) (1)
 Winter ✓
- 3.2. Name the TWO anti-cyclones present at A and B respectively. (2x1) (2)
 SOUTH Atlantic Anti and SOUTH Indian Anti (if no 'South' then it is WRONG – this is how we mark at matric marking) ✓✓
- 3.3. C refers to a coastal low pressure along the coast of South Africa. Explain why this disturbance travels east along the coast. (1x2) (2)
 There is warmer water as it moves east. ✓✓
 Cannot return to cold waters west. ✓✓
- 3.4. Coastal lows often result in berg wind conditions developing. Explain how berg wings develop. (2x2) (4)
 [Accept reasonable answers]
 Air travels from high to low pressure, therefore going from the berg to the coast ✓✓
 HP conditions inland, LP conditions out to sea therefore air moves that way. ✓✓
 Air descends down the mountain heating up, creating berg wind (mention mountain somewhere) ✓✓

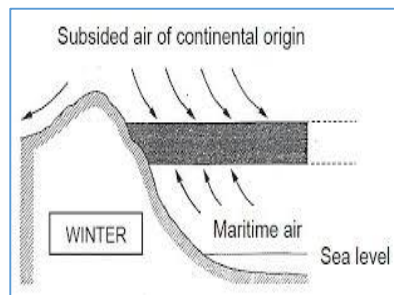
- 3.5. Why are berg winds dangerous? Discuss ONE reason. (1x2) (2)

[Any ONE – Must be a discussion on why it is dangerous. I.e. “it is a hot wind” is not good enough, we need to know WHY this hot wind is dangerous].

Heat dries out crops ✓✓

Veld fires are a possibility with dry conditions ✓✓

- 3.6. Draw a simple diagram explaining how the inversion later present during winter prevents moisture from entering the interior. (3x1) (3)



✓✓✓

- 3.7. What is the name of the anti-cyclone that dominates the interior of South Africa in winter? (1x1) (1)

Kalahari HP ✓

Question 4 – Urban Climates [15]

Read the article below and answer the questions that follow.

Cool solutions for a hotter climate: Tackling urban heat island effect with innovation

Source(s): World Bank, the

Cities worldwide are grappling with the adverse effects of Urban Heat Island (UHI), intensified by rising global temperatures from climate change. The consequences of the UHI effect extend far beyond discomfort, with profound health and economic implications for cities and their residents.

Low-income neighbourhoods, communities lacking green spaces, and other marginalized populations bear a disproportionate burden of the economic costs associated with urban heat. Around 20% of total heat-related excess deaths globally between 2000 and 2019 occurred in East Asia, resulting in over 101,000 fatalities. Moreover, the UHI effect results in decreased productivity and higher expenses for cooling buildings, contributing to an increase in carbon emissions.

The world stands at a climate inflection point where collective action and local interventions are key. To mitigate temperature rises due to climate change, longer-term reductions in carbon emissions and international cooperation are warranted. However, UHI effects due to heat absorption by urban infrastructure and heat production by vehicles and appliances can be mitigated by local interventions. Across cities, there is an urgent need to develop sustainable solutions for mitigation and adaptation in response to UHI effects.

4.1. What problem is being referred to in the article? (1x1) (1)

Urban Heat Islands ✓

4.2. Which area would likely experience the more severe conditions, the CBD or a suburban residential area? (1x1) (1)

CBD ✓

4.3. List THREE factors that contribute to the problem referred to in QUESTION 4.1. (3x1) (3)

[Any THREE]

Artificial surfaces

Artificial sources of heat

Dark surfaces

Lack of green spaces

Lack of environmental awareness/education ✓✓✓

4.4. “Low-income neighbourhoods, communities lacking green spaces, and other marginalized populations bear a disproportionate burden of the economic costs associated with urban heat.” Explain this statement in your own words. (1x2) (2)

[Accept reasonable answers - NEEDS to link to ECONOMIC impacts]

Poor and vulnerable people are less able to afford air-conditioning, they are more likely to get sick as a result, the costs of health and heat are unaffordable to these people, they have the worst jobs for heat etc. ✓✓

- 4.5. In winter urban areas typically suffer from pollution domes. In a short paragraph, briefly discuss TWO economic consequences that pollution domes have on a city and TWO viable solutions that can be implemented to mitigate these problems. (4x2) (8)

[Accept reasonable answers – MUST have both]

ECONOMIC CONSEQUENCES

Costly to clean air

Makes people sick – sick people don't work

Govt hospitals incur costs

Less tourism – less money ✓✓✓✓

SOLUTIONS

Greening

Laws

Fine companies for pollution

Encourage electric cars, public transport etc. ✓✓✓✓

GRAND TOTAL: 60 MARKS